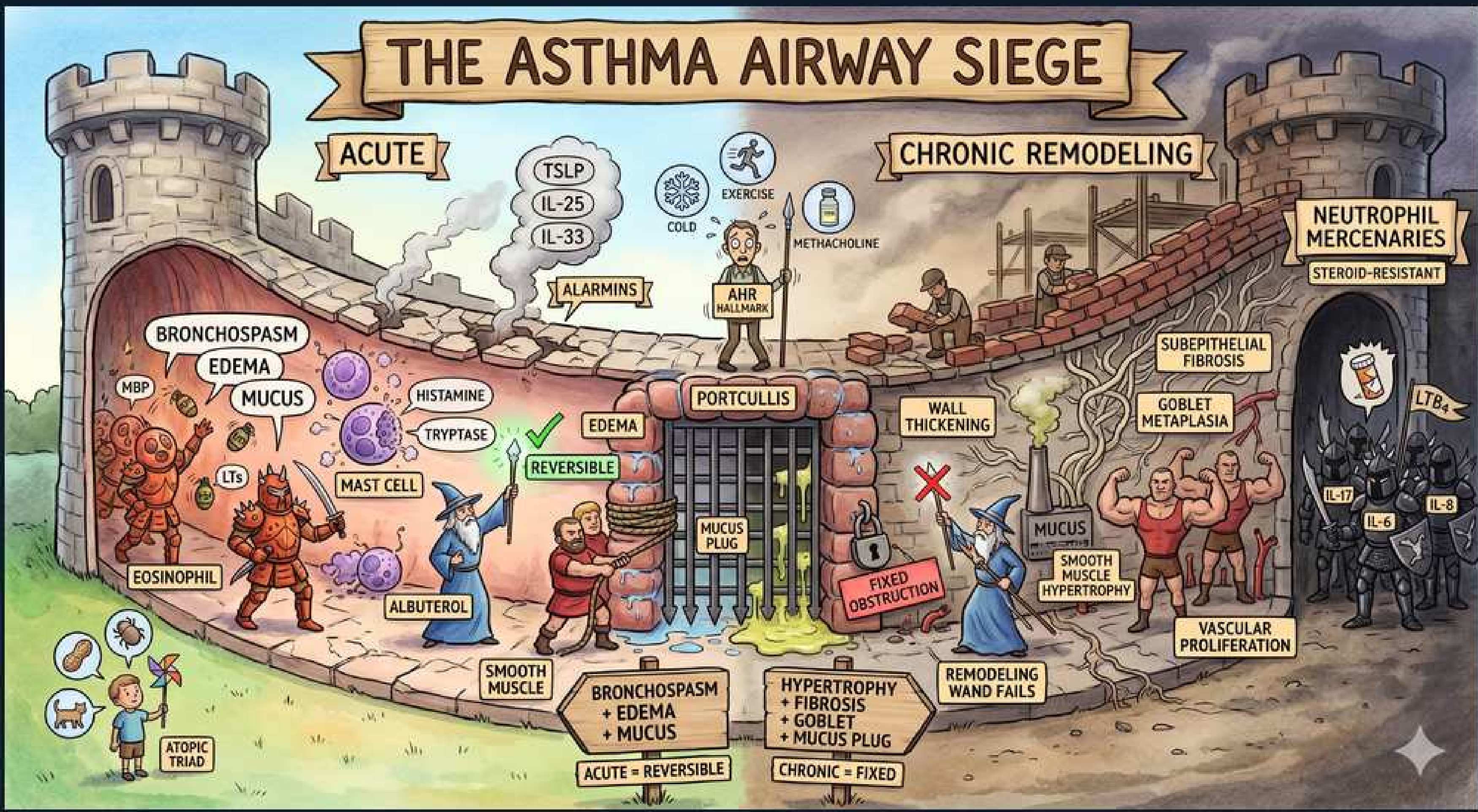
Asthma — The Asthma Airway Siege (pathophysiology)



1. Alarmins TSLP/IL-25/IL-33

WHAT YOU SEE
Bubbles 'TSLP, IL-25, IL-33 → alarmins' rising as smoke.

WHAT IT ENCODES
When airway epithelium is damaged by allergens, viruses, smoke, or pollutants it releases the 'alarmin' cytokines TSLP, IL-25, and IL-33. These activate dendritic cells, Th2 cells, and ILC2s, kick-starting the entire type-2 inflammatory cascade (IL-4/IL-5/IL-13) — they sit at the very top of the pathway.

BOARD TRAP
TSLP is the MOST UPSTREAM druggable target — tezepelumab (anti-TSLP) blocks the alarmin signal and therefore works across phenotypes, including low-eosinophil/low-FeNO and non-type-2 asthma where other biologics fail.

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2. AHR triggers (cold/exercise/methacholine)

WHAT YOU SEE
Icons for cold, exercise, and methacholine feeding 'AHR'.

WHAT IT ENCODES
Airway hyperresponsiveness (AHR) is exaggerated bronchoconstriction to nonspecific stimuli — cold air, exercise, methacholine, histamine, dust. It is quantified by the methacholine challenge (PC20/PD20) and reflects the twitchy, inflamed smooth muscle of the asthmatic airway.

BOARD TRAP
AHR is one of the THREE core features of asthma — variable airflow obstruction, airway hyperresponsiveness, and chronic airway inflammation — and AHR is what bronchoprovocation testing measures when spirometry is normal.

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3. Acute = bronchospasm + edema + mucus

WHAT YOU SEE
Labels 'bronchospasm, edema, mucus' on the acute (left) wall.

WHAT IT ENCODES
Acute obstruction has three reversible components: smooth-muscle BRONCHOSPASM (early, IgE/mast-cell driven), mucosal EDEMA (vascular leak from the late inflammatory phase), and MUCUS hypersecretion/plugging. These dominate the early- and late-phase asthmatic responses.

BOARD TRAP
Acute obstruction is REVERSIBLE — bronchodilator responsiveness (FEV1 ↑≥12% & ≥200 mL) is what distinguishes asthma from the fixed obstruction of COPD.

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4. Mast cell (histamine/tryptase)

WHAT YOU SEE
A mast-cell wizard releasing 'histamine, tryptase'.

WHAT IT ENCODES
IgE-sensitized mast cells degranulate on allergen cross-linking, releasing preformed histamine and tryptase and newly synthesized leukotrienes and prostaglandin D2. This drives the EARLY-phase bronchospasm (minutes) and recruits the cells of the late phase. Serum tryptase is a marker of mast-cell activation (e.g., anaphylaxis).

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5. Eosinophil (MBP)

WHAT YOU SEE
An eosinophil knight with 'MBP' near leukotrienes.

WHAT IT ENCODES
Eosinophils are the hallmark effector cells of type-2 asthma (recruited and sustained by IL-5). They release major basic protein (MBP), eosinophil cationic protein, leukotrienes, and reactive oxygen species that damage epithelium and perpetuate the LATE-phase inflammation. Sputum/blood eosinophilia guides anti-IL-5 biologic therapy.

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6. Albuterol relieves edema (reversible)

WHAT YOU SEE
An 'ALBUTEROL' defender with 'reversible / edema' green check.

WHAT IT ENCODES
Albuterol/salbutamol (SABA) is a beta-2 agonist that raises smooth-muscle cAMP to relax the bronchi, rapidly reversing the bronchospastic component of an acute attack. It is a reliever only — it does nothing for the underlying inflammation that causes edema and remodeling.

BOARD TRAP
SABA relieves spasm but not inflammation — SABA-only treatment is no longer acceptable in GINA; even mild asthma needs ICS (as-needed ICS-formoterol), because reliance on SABA alone increases exacerbations and mortality.

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7. Mucus plug / fixed obstruction portcullis

WHAT YOU SEE
A portcullis gate clogged with a 'mucus plug → fixed obstruction'.

WHAT IT ENCODES
Tenacious mucus plugs (containing eosinophils, Charcot-Leyden crystals, and Curschmann's spirals) physically occlude airways. Over years, recurrent plugging plus remodeling contributes to a partially FIXED obstruction that no longer fully reverses with bronchodilators.

BOARD TRAP
Long-standing, poorly controlled asthma can develop a fixed obstructive component (incomplete reversibility), blurring the line with COPD — this is 'asthma-COPD overlap.'

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8. Wall thickening / remodeling

WHAT YOU SEE
Workers 'wall thickening' under 'chronic remodeling'.

WHAT IT ENCODES
Persistent type-2 inflammation drives airway-wall thickening: smooth-muscle hypertrophy/hyperplasia, subepithelial fibrosis, and mucous-gland expansion. This narrows the lumen structurally and produces the irreversible, fixed component of chronic asthma.

BOARD TRAP
Remodeling is the IRREVERSIBLE part of asthma — early and sustained controller (ICS) therapy aims to prevent it; once established it does not reverse with bronchodilators.

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9. Subepithelial fibrosis

WHAT YOU SEE

Scaffolding 'subepithelial fibrosis' on the chronic wall.

WHAT IT ENCODES

Deposition of collagen beneath the epithelial basement membrane (subepithelial/reticular basement-membrane thickening) by activated myofibroblasts is a defining histologic hallmark of asthma remodeling, present even in mild disease.

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10. Goblet metaplasia + smooth-muscle hypertrophy

WHAT YOU SEE

Labels 'goblet metaplasia', 'smooth muscle hypertrophy', 'vascular proliferation'.

WHAT IT ENCODES

Remodeling features also include goblet-cell hyperplasia/metaplasia (excess mucus, largely IL-13 driven), airway smooth-muscle hypertrophy and hyperplasia (the main contributor to wall thickening and AHR), and angiogenesis/vascular proliferation in the airway wall.

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11. Neutrophil mercenaries (steroid-resistant)

WHAT YOU SEE

Dark 'neutrophil mercenaries — steroid-resistant' soldiers with IL-8/IL-17.

WHAT IT ENCODES

Non-type-2 (neutrophilic) asthma is driven by the Th17 axis — IL-17 and IL-8 (CXCL8) recruit neutrophils rather than eosinophils. It is associated with obesity, smoking, infection, severe/late-onset disease, and responds POORLY to corticosteroids.

BOARD TRAP

Neutrophilic asthma is STEROID-RESISTANT, and the type-2-targeted biologics (anti-IgE, anti-IL-5, anti-IL-4Rα) do NOT help it — tezepelumab (anti-TSLP, upstream) is the broadest option that may benefit low-type-2 disease.

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12. Atopic triad

WHAT YOU SEE

A child labeled 'atopic triad' at the wall base.

WHAT IT ENCODES

Allergic asthma clusters with the other atopic conditions — atopic dermatitis (eczema) and allergic rhinitis/conjunctivitis — in the 'atopic triad/march.' A personal or family history of atopy raises asthma probability and points to the IgE-mediated, allergic phenotype.
